



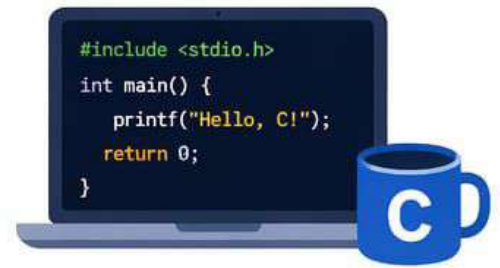
C ROADMAP

Step by Step to Become a C Programmer



Learn. Practice. Build. Master C.

Strong basics today, powerful programs tomorrow.



1



1. Getting Started

- Introduction to C
- Features and Applications
- Setup IDE (VS Code / Code::Blocks / Dev-C++)
- Your First C Program

1-2 Days

OUTCOME

Write and run your first C program.



2



2. Basics of C

- Data Types, Variables, Constants
- Input / Output (printf, scanf)
- Operators and Expressions
- Type Casting

2-3 Days

OUTCOME

Understand the basic building blocks of C.



3



3. Control Flow

- if, else, else if
- Switch Case
- For, While, Do-While Loops
- Break, Continue, Goto

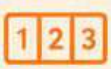
3-4 Days

OUTCOME

Control the flow of your programs.



4



4. Arrays & Strings

- One-dimensional Arrays
- Multidimensional Arrays
- Strings and String Functions
- Character Handling

3-4 Days

OUTCOME

Work with collections of data efficiently.



5



5. Functions

- Define and Call Functions
- Function Parameters and Return Type
- Recursion
- Storage Classes (auto, static, extern, register)

3-4 Days

OUTCOME

Write modular and reusable code.



6



6. Pointers

- What is Pointer?
- Pointer and Variables
- Pointer Arithmetic
- Arrays and Pointers

4-5 Days

OUTCOME

Understand memory and write efficient code.



7



7. Structures & Unions

- Structures
- Arrays of Structures
- Unions
- Typedef

3-4 Days

OUTCOME

Group related data using structures.



8



8. File Handling

- File Operations (fopen, fclose)
- Read / Write Functions
- Text vs Binary Files
- File Pointers

3-4 Days

OUTCOME

Store and manage data in files.



9



9. Advanced Concepts

- Dynamic Memory Allocation (malloc, calloc, realloc, free)
- Command Line Arguments
- Preprocessor Directives (#define, #include, #if, etc.)
- Bitwise Operators

4-5 Days

OUTCOME

Write optimized and powerful C programs.



10



10. Practice & Build Projects

- Solve programming problems
- Build mini projects
- Work on real-world applications

Ongoing

OUTCOME

Build strong logic and real-world experience.



RECOMMENDED TOOLS



VS Code



Code::Blocks



Dev-C++



GitHub

PROJECT IDEAS



Calculator



Student Management System



Library Management System



Quiz Game



File Compressor (with C)



TIPS FOR SUCCESS



Practice every day



Code small, think big



Read and understand code



Solve problems on paper



Be consistent and patient



Consistency is the key to mastery. Keep coding, keep growing!

